

KRISTOFER TSAI

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EDUCATION

University of California, Los Angeles

September 2024 – June 2028

Bachelor of Science in Mechanical Engineering | GPA: 3.8/4.0

Selected Coursework: Dynamics, Fluid Mechanics, Engineering Thermodynamics, Strength of Materials, Engineering Mathematics

MOTORSPORT TECHNICAL PROJECTS

Bruin Formula Racing | Los Angeles, California

March 2026 – Present

Vehicle Dynamics Analyst | Formula Student

- Built BFR's first telemetry baseline in Python by cleaning timestamps and selecting reliable signals from 347 channels.
- Defined telemetry requirements with the data acquisition team and diagnosed sensor, calibration and scaling issues.
- Sign-corrected and resampled inverter data at 10 Hz, validating torque command against feedback to 0.17 Nm MAE.
- Mapped motor speed, torque and electrical power, finding most endurance running occurred at 1,500–3,000 rpm.

Formula 1 Race Strategy Analysis | Los Angeles, California

June 2026 – Present

Independent Project

- Predicted Bahrain 2024 constructor race pace from FP2 long runs, achieving 0.25 s/lap MAE and 91.1% pairwise accuracy.
- Modelled Mercedes' Singapore 2023 VSC stop using pit loss and tyre offset, deriving a 0.98–1.14 s/lap break-even target.
- Tested Verstappen's Brazil 2024 stay-out decision against an earlier stop, isolating the red flag's track-position benefit.
- Modelled Verstappen's France 2021 one- and two-stop options using degradation and catch rate to find the crossover.

PROFESSIONAL EXPERIENCE

GovTech | Singapore

January 2024 – June 2024

Cybersecurity/AI Intern

- Investigated LLM cybersecurity threats and defences, presenting findings to 50+ policymakers and engineers.
- Developed Singapore's first AI Playbook, adopted by 5+ agencies to standardise national AI deployments.
- Drafted technical memos on AI risk scenarios for 200+ government engineers and policy staff.

Singapore Armed Forces | Singapore

January 2022 – November 2023

Soldier

- Served as Supply Assistant at Pasir Laba Camp, coordinating logistics for 500+ personnel during national service.
- Maintained and repaired military equipment in Specialist Cadet Schools, ensuring readiness for 1,000+ cadets.

RESEARCH & ANALYTICS EXPERIENCE

rideOS | San Francisco, California

May 2020 – January 2021

Researcher

- Conducted machine-learning research on NBA game prediction under the supervision of a rideOS software engineer.
- Processed 1,200+ NBA games in Python and engineered 15+ features for predictive modelling.
- Evaluated six classifiers, including Naive Bayes and J48, using F-score to compare predictive performance.
- Achieved a maximum F-score of 0.61, outperforming baseline models by 12%.

Bruin Sports Analytics | Los Angeles

September 2024 – Present

Data Analyst

- Analysed PGA Tour data to quantify the relationship between driving distance and scoring performance.
- Found driving distance explained little variation in scoring average after outlier refinement, with an R^2 of 0.02.
- Presented findings to 50+ attendees and published the study through [Bruin Sports website](#).

DataRes | Los Angeles

January 2025 – Present

Data Analyst

- Analysed 50,000+ dating-app reviews with VADER NLP, revealing a 25% COVID-era fall in positive sentiment.
- Built Matplotlib visualisations and presented consumer-behaviour findings to 100+ peers and faculty.
- Cleaned review text and compared sentiment trends across pre- and post-COVID periods using Python.

SKILLS & INTERESTS

Programming & Data: Python (Pandas, NumPy, Matplotlib, FastF1), OpenF1 API, MATLAB, C++, R, SQL

Engineering & Tools: SolidWorks, Telemetry Analysis, Data Validation, Git/GitHub

Activities: UCLA Club Golf, UCLA Club Tennis, Formula 1, High-Performance Driving (HPDE)